

Generative Adversarial Networks (GANs): A Survey of Theoretical Models, Applications, and Development Trends with a Focus on Parallel Intelligence

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Abstract – This paper discusses Generative Adversarial Networks (GANs), which are a type of generative model introduced in 2014. GANs consist of a generator and a discriminator that learn simultaneously through a minimax game process. The goal of GANs is to reach a Nash equilibrium, where the generator captures the distribution of real samples. GANs have become a popular research topic in the field of artificial intelligence, with applications in image and vision, speech and language processing, malware detection, and chess game programs. The paper surveys the state of the art of GANs, including their theoretical and implementation models, typical applications, advantages and disadvantages, and development trends. The relation between GANs and parallel intelligence is also investigated. In order to assist researchers in identifying fresh difficulties and chances in this fascinating field, we also identify future research directions in the area of GAN-based image synthesis.

Keywords – GANs, Nash equilibrium, parallel intelligence

I. INTRODUCTION

A Generative Adversarial Network (GAN) is a deep learning model composed of two neural networks: a generator and a discriminator. The process involves training a generator network to produce new images that are similar to a set of training images. The generator is trained in conjunction with a discriminator network, which is

trained to distinguish between generated images and the real images from the dataset. The generator network typically consists of a series of transposed convolutional layers, which increase the spatial resolution of the input noise, until a final image is produced. The discriminator network, on the other hand, is a series of convolutional layers, which decrease the spatial resolution of the input image, and produce a single output value representing the probability that the input image is real. The generator creates new data samples, while the discriminator evaluates the authenticity of the generated samples. GANs have been successful in various applications such as image synthesis, style transfer, and anomaly detection. During the training process, the generator produces images, which are then evaluated by the discriminator. The generator is updated based on the feedback from the discriminator, and the process continues until the generator produces images that are indistinguishable from the real images by the discriminator. Once trained, the generator can be used to produce new images by inputting random noise. These generated images can be used for a variety of tasks such as image synthesis, style transfer, and data augmentation. GANs are particularly useful for generating high-quality images, since the generator learns to produce images that are similar to the real images from the training set. The task is to design and train a Generative Adversarial Network (GAN) that can generate new images that are similar to a given set of training images. The generated images should be of high-quality and should be indistinguishable from real images by a discriminator network. The

GAN should be able to handle various types of images, such as natural images, artistic images, and medical images. Additionally, the GAN should be able to generate images of different resolutions and should be able to handle images with different levels of complexity. The training process should be efficient and the GAN should be able to generalize well to unseen data.

1.1 Main reasons for emergence of GAN

A. Boom of Artificial Intelligence

The five levels of artificial intelligence (AI), including discrimination, generation, understanding, and creation and application. The authors argue that these levels are interrelated and reinforce each other, but also have big gaps between them that require new breakthroughs. The paper suggests that understanding is a common topic in both the two stages of AI perception and cognition, and the four levels of AI. Although measuring understanding in AI is difficult, the ability of machines to create can reflect their level of understanding. The paper also suggests that generative models such as GANs can deepen AI research by their ability to generate data samples and reflect understanding to a certain degree.

B. Accumulation of Generative Models

The importance of generative models in the AI field and their two research perspectives: "humans understand data" and "machines understand data." Generative models are obtained by learning with generative methods and involve distribution hypothesis and parameter estimation. From the perspective of "humans understand data," the generative model has a distribution that humans can understand but may have limitations for learning machines. From the perspective of "machines understand data," the generative model generates data samples from the distribution without explicit hypothesis and is less interpretable to humans. The limitations in model training and data generation have been a barrier to realizing the four levels of AI, and the article suggests designing a new paradigm of generative models to break through these barriers.

C. Deepening of Neural Networks

Neural networks have been revived due to the increase in computing power and data scale. They have been successfully used to solve complex nonlinear problems, such as image classification,

speech recognition, and natural language processing. The success of neural networks is due to their model characteristics, such as the general backpropagation algorithm for training, flexible structural design, and theoretical capacity to approximate any function. With the development of computing devices, neural networks with more parameters can be trained faster, further promoting their popularity.

D. Success of Adversarial Idea

N Adversarial idea is a concept that has been successfully introduced in machine learning and AI. Game-theoretic machine learning combines game theory with machine learning to model human dynamic strategy and optimize advertisement auction mechanisms. AlphaGo, a famous AI program that defeated human masters in the board game Go, used two networks to compete in the training process. This encouraged the hidden nodes in the network to be statistically independent and served as a regularization factor in the training process. Adversarial idea has also been used to train neural networks with domain adaptation, where the feature generator transforms the source-domain data and the target-domain data into high-level abstract features, making the domains of features difficult to discriminate as far as possible. Adversarial samples are another concept that contains the adversarial idea. These refer to samples that are classified into the wrong category or classified into a real category with high confidence but have a big difference from the real samples. Researchers have used adversarial learning idea to generate positive examples that are hard for object detectors to recognize, thereby making the detectors more robust to occlusion and deformation. Adversarial learning, adversarial networks, and adversarial samples all contain adversarial idea but have different objectives. These achievements have inspired more researchers to explore Generative Adversarial Networks (GANs).

II. ARCHITECTURE AND PRINCIPLE

The GAN architecture consists of two neural networks: a generator and a discriminator. The generator network takes a random noise vector as input and generates synthetic data that resembles the real data. The discriminator network takes both real and synthetic data as input and predicts

whether the input is real or synthetic. The generator learns to generate plausible data. The generated instances become negative training examples for the discriminator. The discriminator learns to distinguish the generator's fake data from real data. The discriminator penalizes the generator for producing implausible results.

The GAN architecture consists of a generator network and a discriminator network. The generator network takes a random noise vector as input and produces a sample that is similar to the real data. The discriminator network takes a sample from either the real data or the generator and produces a probability score that indicates how likely the sample came from the real data. During training, the generator tries to generate samples that can fool the discriminator, while the discriminator tries to correctly distinguish between the real and generated samples.

The training of GANs is an adversarial optimization process, where the two networks are trained in a back-and-forth manner. The generator tries to minimize the discriminator's ability to distinguish between the real and generated samples, while the discriminator tries to maximize its ability to distinguish between them. The training process continues until the generator produces samples that are indistinguishable from the real data, and the discriminator can no longer distinguish between them. At this point, the Nash equilibrium is reached, and the generator has learned to generate samples that follow the same distribution as the real data.

The idea behind GANs (Generative Adversarial Networks) is based on the concept of Nash equilibrium from game theory. GANs consist of two players, a generator and a discriminator, who engage in a two-player minimax game. The generator tries to generate samples that are similar to the real data, while the discriminator tries to distinguish between the real and generated samples. The two players try to improve their strategies to win the game, and the equilibrium point of the game corresponds to the generator generating samples that are indistinguishable from the real data.

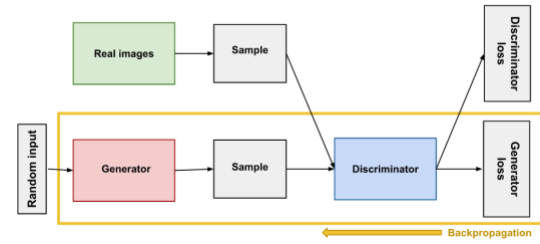


Figure 1

III. TYPES OF GANs

1. Wasserstein GAN (W-GAN)

is a variant of Generative Adversarial Networks (GAN) that aims to overcome some of the limitations of traditional GANs. The main problem with GANs is that they can be difficult to train, and there is no guarantee that the generated samples will be of high quality.

W-GAN addresses these issues by introducing a new loss function that measures the Wasserstein distance between the real and fake data distributions. This loss function, known as the Earth Mover's Distance (EMD), is a more meaningful measure of the distance between distributions than the Jensen-Shannon Divergence used in traditional GANs.

$$W(\mathbb{P}_r, \mathbb{P}_g) = \inf_{\gamma \in \Pi(\mathbb{P}_r, \mathbb{P}_g)} \mathbb{E}_{(x,y) \sim \gamma} [\|x - y\|] ,$$

By using the EMD, W-GAN is able to provide more stable and meaningful gradients during training, making it easier to train and producing better-quality samples. W-GAN has been shown to be effective in a variety of applications, including image generation, style transfer, and super-resolution.

2. Least Squares GAN (LS GAN)

is a type of generative adversarial network (GAN) that was introduced by Xudong Mao, et al. in 2017. It addresses a problem in traditional GANs, where the generator and discriminator can reach a point where they no longer improve, and the training process becomes unstable.

In LS-GAN, the loss function is modified to use the least squares method, which leads to more stable training and better results. Specifically, the generator and discriminator are trained to minimize the mean squared error (MSE) instead of the binary cross-entropy loss used in traditional GANs. This modification provides several advantages, such as increased stability, faster convergence, and better image quality.

LS-GAN has been shown to be effective in various applications, including image generation, text-to-image synthesis, and video prediction. It is an important development in the field of GANs and has led to significant advances in generative modeling.

3. *Semi Supervised GAN (Semi GAN)*

is a type of generative adversarial network (GAN) that combines unsupervised and supervised learning for improved performance on semi-supervised tasks. It was introduced by Augustus Odena, et al. in 2016.

In Semi-GAN, the discriminator is trained to not only distinguish between real and fake data but also to classify data into different categories. The generator is then trained to generate data that is not only realistic but also belongs to the correct category. This semi-supervised approach allows the model to learn from a small amount of labeled data, while also making use of a large amount of unlabeled data.

Semi-GAN has been applied to various tasks such as image classification, object recognition, and natural language processing. It has been shown to achieve state-of-the-art results in semi-supervised learning, surpassing other methods that rely solely on supervised learning or unsupervised learning. Overall, Semi-GAN is an important development in the field of GANs and has the potential to improve the performance of machine learning models in scenarios where only a limited amount of labeled data is available.

4. *Conditional GAN (C-GAN)*

is a type of generative adversarial network (GAN) that has been modified to allow for conditional image generation. It uses a two-part architecture consisting of a generator and a discriminator, which work in opposition to each other. The generator takes a random noise vector and a conditional label as input and generates a corresponding image. The discriminator receives both the generated image and the corresponding label, and its job is to determine whether the image is real or fake. C-GAN is a powerful tool for generating images that meet specific criteria or have certain characteristics. For example, it can be used to generate images of specific objects, such as cars or

flowers, or to generate images with particular attributes, such as images of people with blonde hair or green eyes. C-GAN can be used in a variety of applications, including computer graphics, video game development, and image manipulation.

5. *Bidirectional GAN (BiGAN)*

is a variant of GANs (Generative Adversarial Networks) that introduces an encoder network, in addition to the generator and discriminator networks, in order to learn a mapping between the latent space and the input data space. This means that instead of generating data from random noise, BiGAN generates data by encoding real data into the latent space, and then decoding it back into the input data space.

The BiGAN architecture is similar to that of GANs, with the difference being the addition of the encoder network. The generator network takes a random noise vector and generates a synthetic sample, while the discriminator network attempts to distinguish between the synthetic samples and real ones. The encoder network maps the real data back to the latent space, which is then fed into the generator network to generate synthetic samples. The encoder in BiGAN can also be used for tasks such as feature extraction and representation learning. Since it learns a mapping from the input data space to the latent space, it can be used to extract useful features from the input data that can be used for downstream tasks such as classification. BiGANs have been used for a variety of applications, including image synthesis, image manipulation, and anomaly detection. They have shown promising results and continue to be an active area of research in the field of generative modeling.

6. *Information Maximising GAN (InfoGAN)*

is a type of Generative Adversarial Network (GAN) that can learn and disentangle meaningful and interpretable features from input data. Unlike traditional GANs, InfoGAN includes a mechanism that maximizes the mutual information between some subset of the generator's input noise and the generated samples. By doing so, the model can discover and learn to control specific aspects of the generated data, such as object orientation, shape, and style.

InfoGAN is based on the idea that the generator network should not only learn to generate realistic samples but also learn to generate diverse samples

with different features. The goal of InfoGAN is to learn a factorized representation of the input data that is both compact and informative.

One of the main benefits of InfoGAN is that it can generate samples with high variability while maintaining a consistent style, making it useful for applications such as image synthesis and data augmentation. Additionally, InfoGAN has been used for unsupervised representation learning and has achieved state-of-the-art results on several benchmark datasets.

InfoGAN is a powerful variant of GANs that allows for disentangling and controlling the features of generated data by maximizing the mutual information between the generator's input noise and the generated samples.

7. Auxiliary Classifier GAN (ACGAN)

is a type of Generative Adversarial Network (GAN) that can generate high-quality images with a given label or class. Unlike traditional GANs, ACGAN includes an auxiliary classifier in the discriminator network that helps to control the generated samples' class labels.

The goal of ACGAN is to generate realistic images that not only look like real samples but also belong to a specific class. ACGAN uses the generator network to generate samples and the discriminator network to classify them into one of the pre-defined classes.

One of the significant benefits of ACGAN is that it can generate high-quality samples with specific labels, making it useful for applications such as image synthesis and data augmentation. Additionally, ACGAN has been used for semi-supervised learning, where the generator network can be trained on both labeled and unlabeled data. ACGAN has achieved state-of-the-art results on several benchmark datasets, including CIFAR-10 and MNIST, demonstrating its ability to generate high-quality images with specified class labels. In summary, ACGAN is a powerful variant of GANs that can generate high-quality images with specific class labels by including an auxiliary classifier in the discriminator network.

8. Sequence GAN (SeqGAN)

is a type of Generative Adversarial Network (GAN) that is specifically designed for generating sequences, such as natural language text or music. Unlike traditional GANs, SeqGAN includes a reinforcement learning-based approach to train the

generator network to generate realistic and coherent sequences.

The goal of SeqGAN is to generate sequences that are not only plausible but also coherent and meaningful. SeqGAN uses a generator network to generate sequences and a discriminator network to evaluate the quality of the generated sequences.

The generator is trained using a reinforcement learning approach that considers the output sequence as a sequence of actions, and the discriminator provides a reward signal to the generator based on the quality of the generated sequence.

One of the significant benefits of SeqGAN is that it can generate sequences with improved coherence and diversity, making it useful for applications such as text generation and music generation.

Additionally, SeqGAN has been used for unsupervised text summarization, where the generator network can be trained on unstructured text data to generate a summary of the text. SeqGAN has achieved state-of-the-art results on several benchmark datasets, including text and music datasets, demonstrating its ability to generate high-quality and meaningful sequences.

Different GANs

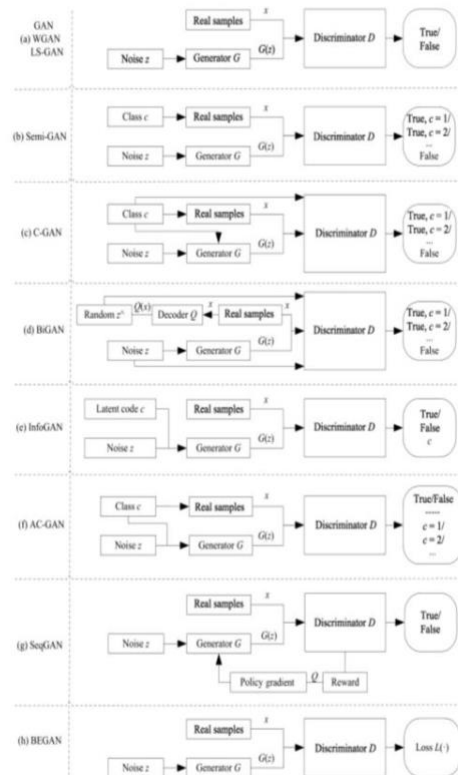


Figure 2

IV. APPLICATIONS OF GAN

A. APPLICATIONS TO IMAGE AND VISION COMPUTING

1. Image Synthesis: GANs can be used to generate high-quality images of objects, scenes, or people that do not exist in the real world. Such images can be used for training deep learning models, creating virtual reality environments, and even for generating visual content for media and advertising.



FIGURE 6. Image super-resolution generated by different GANs. (a) Ground truth; (b) SRGAN; (c) ESRGAN.

Figure 3

2. Image-to-Image Translation: GANs can be used to translate images from one domain to another, such as converting a black and white image to color, enhancing low-resolution images, or transforming a photo into a painting or a sketch.

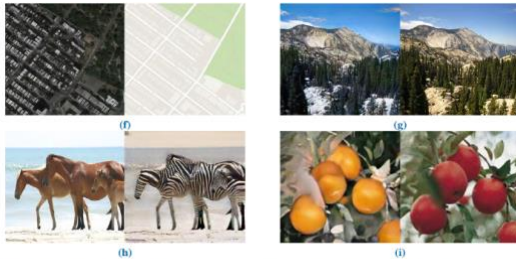


FIGURE 7. Image translation generated by CycleGAN. (a) Original; (b) Van Gogh; (c) Monet; (d) Cezanne; (e) Ukiyoe; (f) Satellite to map; (g) Winter to summer; (h) Horse to zebra; (i) Orange to apple.

Figure 4

3. Super-Resolution: GANs can be used to increase the resolution of images by generating high-resolution images from low-resolution ones. This can be useful in medical imaging, where higher resolution images can provide more accurate diagnoses.
4. Data Augmentation: GANs can be used to generate synthetic data that can be used to augment training data for deep learning models. This can help in improving the accuracy and robustness of models.
5. Image Editing: GANs can be used for interactive image editing, where users can modify specific attributes of an image, such as changing the color of an object, removing or

adding objects, or changing the lighting conditions.

Overall, GANs have tremendous potential in image and vision computing, and their applications are likely to expand as the technology continues to develop.

B. APPLICATIONS TO SPEECH AND LANGUAGE PROCESSING

1. Speech synthesis: : GANs can be used to generate realistic-sounding speech, which can be used in various applications, such as text-to-speech systems, voice assistants, and other human-machine interfaces.
2. Speech enhancement: GANs can be used to remove noise from speech signals, improving the intelligibility and quality of the speech.
3. Speech recognition: GANs can be used to generate synthetic speech that can be used to train speech recognition systems, making them more robust to variations in speaker accent, speaking style, and background noise.
4. Machine translation: GANs can be used to generate synthetic examples of text in a target language, which can be used to train machine translation systems, enabling them to translate more accurately.
5. Sentiment analysis: GANs can be used to generate synthetic text examples with specific sentiment, which can be used to train sentiment analysis systems, making them more accurate and robust.
6. Dialogue generation: GANs can be used to generate synthetic text examples that can be used in conversational agents, such as chatbots and virtual assistants, enabling them to have more natural and engaging conversations with users.

V. RELATION BETWEEN GAN AND PARALLEL INTELLIGENCE

The parallel systems theory and ACP approach for modeling and controlling complex systems, which emphasizes virtual-real interaction. This approach uses artificial systems to represent the actual system, computational experiments to learn and evaluate various computational models, and parallel execution to improve the actual system's performance. The parallel systems theory and ACP approach have evolved into a more generalized parallel intelligence theory. GANs, as a powerful class of effective generative models, deepen the philosophy of parallel systems' virtual-real

interaction and integration, and can merge into the systematic research of parallel intelligence.

1. GANs and parallel vision: Parallel vision is an extension of the ACP approach in the field of vision computing, aiming to establish systematic theories and methods for visual perception and understanding of complex scenes. It uses computer graphics, virtual reality, machine learning, and knowledge automation to construct artificial scenes, utilize computational experiments to design and evaluate vision algorithms, and apply parallel execution to online optimize the vision system. GANs can generate large-scale diversified image datasets in parallel vision, which can be combined with real datasets to train vision models and improve their generalization ability.
2. GANs and parallel control: Parallel control is a specific application of the ACP approach in complex system control, using artificial systems for modeling, computational experiments for analysis and evaluation, and parallel execution for control and management. It is an extension of feedback control for dealing with engineering and social complexities. GANs can be used in parallel control to conduct predictive and feedback learning of artificial and actual systems, as well as to enable imitation and reinforcement learning of the control unit.

GANs and parallel learning: Parallel learning

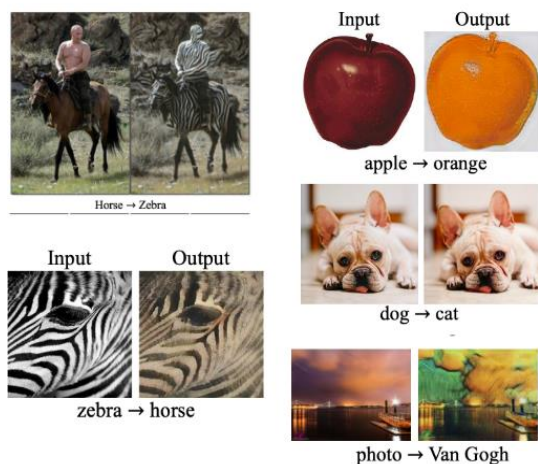


Figure 5

is an extension of the ACP theory in the learning field, with a focus on using predictive learning to explore time-varying data, ensemble learning to explore spatial-distributed data, and prescriptive learning to explore the direction of data generation. It can be seen as a new theoretical framework for machine learning and is related to parallel vision and control. GANs can be combined with parallel learning for big data generation and predictive learning through computational experiments.



Figure 6

VI. LIMITATIONS

Although GANs have helped solve many problems for generative models and have inspired other AI methods, they still have limitations. GANs use adversarial learning, but it hasn't been proven yet that the model will converge and reach an equilibrium point. The training process needs to balance and synchronize two networks, but this can be difficult, leading to unstable training and a common issue of poor interpretability. GANs also have a problem with mode collapse, where generated images have little difference for human understanding. However, researchers are continuously developing techniques to reduce these limitations. Wasserstein GAN, for example, overcomes the training instability problem and partially solves the mode collapse problem. Further research is needed to completely avoid mode collapse and optimize the training process. In addition to improving GANs, research is also focused on generating a variety of data that can interact with humans and integrating GANs with other AI methods to develop new applications. Ultimately, using GANs to enhance the ability of AI to understand the world and stimulate its creativity is an important area of research. Although our strategy frequently produces persuasive results, the outcomes are not always favourable. Several common failure scenarios are seen in Figure 5. The method frequently works on translation assignments involving colour and texture changes, like many of those mentioned above. We have unsuccessfully investigated

problems that demand geometric modifications. For instance, the learnt translation degenerates to making minor adjustments to the input on the job of dog-to-cat transformation (Figure 5). This might be a result of the architecture decisions we made for our generator, which are designed to operate well with aesthetic changes. Future studies must address the difficulty of handling more complex

and radical transformations, notably geometrical alterations.

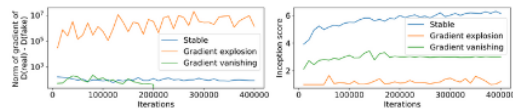


Figure 7

The distribution properties of the training datasets are to blame for some failure occurrences. For instance, the horse and zebra example in Figure 5 to the right has been perplexed since our model was trained on ImageNet's wild horse and zebra synsets, which do not contain any pictures of people riding horses or zebras. We also see a persistent difference between the outcomes of our unpaired technique and those obtained using paired training data. This gap may be difficult, if not impossible, to close in some circumstances. For instance, our method occasionally reorders the labels for the building and tree in the output of the photos-labels task. It could be necessary to use a weak type of semantic supervision to clear up this ambiguity. Incorporating weak or semi-supervised data could provide significantly more potent translators while still only incurring a small portion of the full-supervised systems' annotation costs.

VII. CONCLUSIONS

GANs and genetic algorithms can be used to generate cover images for steganography. GANs are effective for detecting anomalies in aerial images, and f-AnoGAN outperformed other models. SuperstarGAN is an improved version of StarGAN for large-scale domains. GANs have potential for disease classification, tumor detection, brain aging modeling, and disease progression estimation in neuroimaging and clinical neuroscience. GANs can predict the progression of brain lesions and tumor growth, but challenges such as cross-study robustness and generating biologically relevant outputs need to be addressed. GANs have potential for medical imaging analysis, but challenges such as modeling continuous disease

progression, interpreting output, and ensuring algorithmic fairness need to be addressed. GANs have potential for various neuroimaging and clinical neuroscience applications, and future research should focus on algorithm reproducibility, interpretability, and fairness.

IX. DATSETS

1. MNIST dataset is a curated list of all handwritten digits. We have used dataset for quick validation of our model. All images are scaled to 28X28.

2. The CelebA dataset consists of over 10k identities and 200k total images. All images are originally of size 160X160 pixels. They are rescaled to 28X28 pixels.

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